

Diabetes Progression

A Statistical Study Using Multiple Regression in R

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1. Introduction

This analytical report addresses Diabetes Progression over time. The core question focuses on investigating how baseline clinical, physical, and biochemical measurements can assist predicting one-year disease progression in patients diagnosed with diabetes. Diabetes mellitus represents a global healthcare challenge characterized by complex physiological interactions. Understanding which patient health metrics can serve as reliable early indicators of rapid disease advancement or metrics that assist the recognition and controlling the severity of issue is essential for allocating preventive resources, establishing target clinical interventions, and designing management protocols based on data. In this study, we aim to find parameters and features that statistically show signs of connection to the progression of this disease.

2. Core Research Question

How effectively do physical health metrics (specifically Body Mass Index [BMI] and blood pressure) and serum biochemical markers predict quantitative diabetes disease progression over a one year, and does combining physical and biochemical predictors into a multiple linear regression model improve predictive utility compared to single-predictor models significantly?

Subordinate Questions: The study addresses four subordinate analytical objectives:

- 1. What data are available, and are there any important data-quality or data-use issues?
- 2. What are the most important characteristics of the dataset before modelling begins?
- 3. Which variables appear most promising for further analysis, and why?
- 4. What do the simple and multiple regression models tell us about diabetes progression?
- 5. How much confidence should we have in the model results, and what are the main statistical weaknesses or uncertainties?
- 6. What can and cannot reasonably be concluded from this analysis?

3. Data Description, Preparation, and Ethical Considerations

The analysis utilizes the diabetes repository dataset from NCSU , consisting of 442 patient observations collected at baseline, each paired with a measurement of disease progression recorded one year post-baseline. The dataset includes 11 variables:

Variable Name	Description	Data Type	Summary Stat (Mean $\pm$ SD / Range)
age	Patient age in years	Continuous / Numeric	48.52 $\pm$ 13.11 (19 to 79)
sex	Biological sex indicator	Categorical Factor (1=Male, 2=Female)	1.47 $\pm$ 0.50 (46.8% Cat 1, 53.2% Cat 2)
bmi	Body mass index (kg/m ²)	Continuous / Numeric	26.38 $\pm$ 4.42 (18.0 to 42.2)
bp	Mean arterial blood pressure (mmHg)	Continuous / Numeric	94.65 $\pm$ 13.83 (62.0 to 133.0)
s1	Total blood serum cholesterol (tc)	Continuous / Numeric	189.14 $\pm$ 34.61 (97.0 to 301.0)
s2	Low-density lipoproteins (ldl)	Continuous / Numeric	115.44 $\pm$ 30.41 (41.6 to 242.4)
s3	High-density lipoproteins (hdl)	Continuous / Numeric	49.79 $\pm$ 12.93 (22.0 to 99.0)
s4	Total cholesterol / HDL ratio (tch)	Continuous / Numeric	4.07 $\pm$ 1.29 (2.0 to 9.09)
s5	Log serum triglycerides level (ltg)	Continuous / Numeric	4.64 $\pm$ 0.52 (3.26 to 6.11)
s6	Blood glucose level (glu)	Continuous / Numeric	91.26 $\pm$ 11.50 (58.0 to 124.0)
progression (Y)	Quantitative measure of disease progression	Continuous Outcome Variable	152.13 $\pm$ 77.09 (25.0 to 346.0)

Our assessment confirmed 0 missing values across all 442 rows and 0 duplicate entries. Sex was re-encoded as an explicit categorical factor rather than a numeric measurement to prevent erroneous continuous modeling. From an ethical perspective, patient observations were already fully de-identified prior to analysis, ensuring zero risk of personal health information (PHI) disclosure and data-leakage for malintentioned actions. Furthermore, ethical modeling practices require acknowledging that statistical associations must not be misconstrued as clinical definitive diagnoses or direct evidence of treatment efficacy or the actual cause to an effect.

4. Analysis Workflow in R

The study was structured into four phases using R:

Phase 1 — Data Preparation & Cleaning: Tabular data was imported, validated for data integrity (missing values and duplications), and variable names were standardized. The sex variable was explicit factor-encoded.

Phase 2 — Exploratory Data Analysis: Parametric metrics (mean, standard deviation) and non-parametric metrics (median, interquartile range) were calculated. Distributions of progression and physical predictors were plotted using histograms, kernel density estimates, and boxplots (The entire plot can be found in the project repository).

Phase 3 — Bivariate Relationship Inspection: Bivariate correlation matrices (Pearson) were computed across all 10 predictors and the outcome variable. Scatterplots with linear fit lines were generated to assess linearity and heteroscedasticity.

Phase 4 — Regression Modeling & Diagnostics: Linear regression models were fitted using ordinary least squares (OLS). Model 1 evaluated single-predictor performance (progression ~ bmi), while Model 2 incorporated physical and top biochemical predictors (progression ~ bmi + bp + s5). Model diagnostic checks included residual vs. fitted plot analysis and Variance Inflation Factor (VIF) calculation.

5. Key Results

Exploratory distribution analysis revealed that disease progression ranges widely from 25 to 346 with a mean of 152.13 (SD = 77.09) and median of 140.50 (IQR = 124.50), demonstrating slight right-skewness. Body Mass Index averaged 26.38 kg/m² (SD = 4.42, median = 25.70, IQR = 6.08).

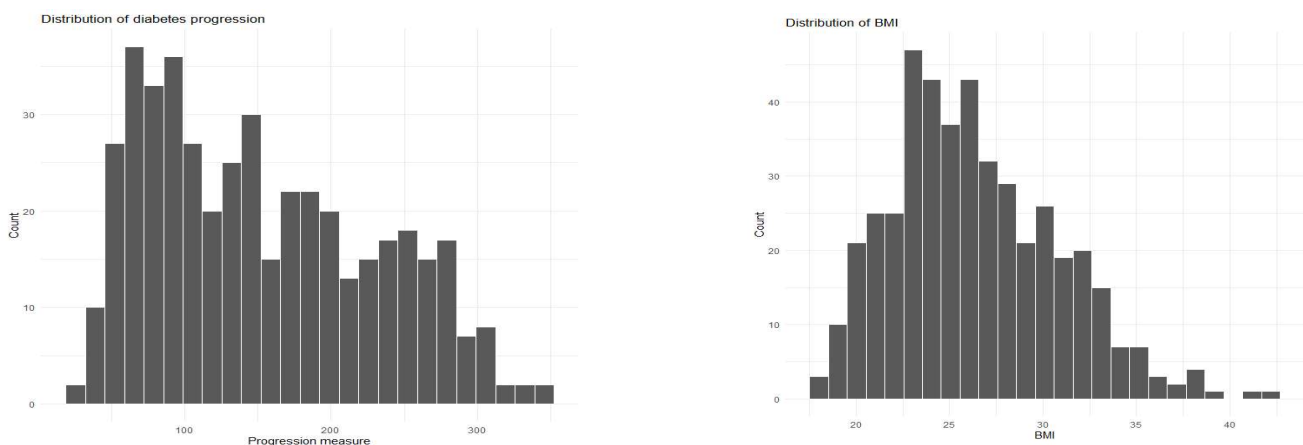


Figure 1: Baseline distribution histograms with density overlays for disease progression (Y) and BMI.

Pearson correlation analysis established that the three predictors most likely capable of helping a predictive model for disease progression are Body Mass Index ($r = 0.586$, $p < 0.001$), log serum triglycerides s5 ($r = 0.566$, $p < 0.001$), and mean blood pressure bp ($r = 0.441$, $p < 0.001$). We must mention that Inter-predictor correlation analysis flagged substantial collinearity between total cholesterol s1 and LDL s2 ($r = 0.897$) as well as HDL s3 and TCH s4 ($r = -0.738$), therefore we include s5 as the primary biochemical representative in multiple regression modeling.

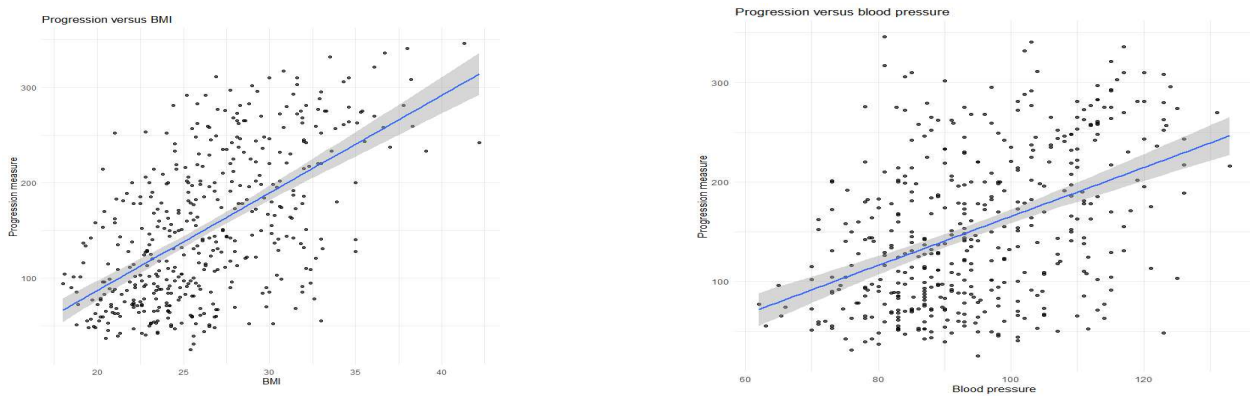


Figure 2: Bivariate scatterplots illustrating positive linear trends for BMI and Blood Pressure against disease progression.

Regression modeling showed some performance progression between the simple linear model and the multiple linear model. Table 3 summarizes the coefficient estimates, standard errors, test statistics, and model fit metrics.

Model Specification	Predictor Variable	Coefficient (β)	Std. Error	t-statistic	p-value	Model R^2	Residual SE
Simple Regression (progression ~ bmi)	Intercept	-117.77	18.02	-6.54	< 0.0001	0.344 (Adj: 0.342)	62.52 units
	bmi	+10.23	0.67	+15.19	< 0.0001		
Multiple Regression (progression ~ bmi + bp + s5)	Intercept	-334.88	25.88	-12.94	< 0.0001	0.480 (Adj: 0.477)	55.78 units
	bmi	+6.50	0.70	+9.32	< 0.0001		
	bp	+0.90	0.22	+4.17	< 0.0001		
	s5	+49.58	5.89	+8.42	< 0.0001		

- **MODEL COMPARISON HIGHLIGHT:** Transitioning from a single-predictor model (BMI alone) to a multiple predictor model (BMI + BP + s5) increased explained disease progression variance from 34.4% to 48.0% ($p < 2e-16$), Residual prediction standard error reduced from 62.52 down to 55.78 units.

6. Interpretation, Statistical Uncertainty, and Limitations

In the multiple linear regression model, the slope coefficients represent controlled marginal effects:

BMI Effect: Holding blood pressure and s5 constant, each 1 kg/m² increase in BMI is associated with an average increase of 6.50 units in disease progression (95% CI: [5.13, 7.87], $p < 0.0001$). This controlled slope is lower than the simple regression slope (+10.23), demonstrating that simple regression inflates the apparent effect of BMI by absorbing variance shared with blood pressure and serum lipids, therefore for multiple reasons linked with the change in the progression value.

BP and Serum Effects: Holding BMI and s5 constant, each 1 mmHg increase in mean blood pressure increases progression by 0.90 units (95% CI: [0.48, 1.33], $p < 0.0001$). Holding physical metrics constant, each 1-unit increase in log serum triglycerides s5 yields a 49.58 unit increase in progression (95% CI: [38.00, 61.15], $p < 0.0001$).

Diagnostic evaluation confirmed the statistical validity of the multiple model. Variance Inflation Factor (VIF) analysis returned low values (BMI = 1.35, BP = 1.27, s5 = 1.34), suggesting that multicollinearity is negligible among the selected predictors. Residual plots ($R^2 = 0.480$) (Figure 3) show that the residuals are quite spread and far from the model's fitted regression line. The shape of the scatters suggest possible non-linearity of the relationship between the features and the target, and error values being statistically dependent. The variance also seems to be non-constant, meaning that the model is not fully accountable for describing the change in the progression at its current state.

Three vital analytical limitations must be highlighted for decision-makers:

- **1. Association vs. Causality:** The underlying data is strictly observational. As a result, these statistical slopes reflect predictive associations, not proven causal mechanics. Medical interventions that artificially alter BMI or BP cannot be assumed to yield direct linear reductions in progression without controlled trials and must be subjected to further tests, improvements, and, ultimately, control of the medical staff.
- **2. Confounding Variables:** Crucial clinical determinants—such as patients' diet, physical activity levels, prescribed medications, genetic background, mental state, and duration of disease prior to baseline—were unmeasured, introducing potential confounding bias.

- 3. Uncertainty:** While the overall model is highly statistically significant ($p < 0.0001$), 52% of total progression variance remains unexplained ($R^2 = 0.480$). The Residual Standard Error of 55.78 units represents great uncertainty relative to the outcome mean (152.13), rendering the model far more suitable for population risk and suggestive alarm utility than point estimation for individual clinical management and standalone risk factors.

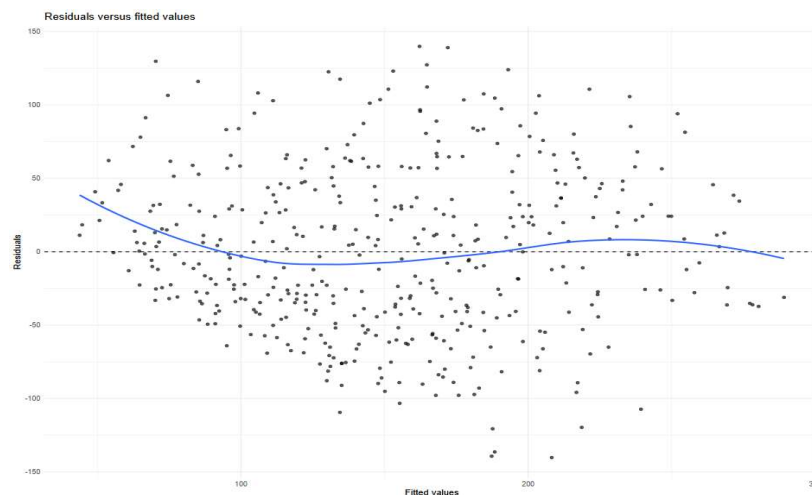


Figure 3: Residuals vs. Fitted Values diagnostic plot for the multiple linear regression model

7. A Summary for Non-Specialists

A patient's body weight (BMI), blood pressure, and serum lipid profile (s5) can function as early warning signals for 1-year diabetes progression across about 450 patients that have been studied. Relying on body weight alone misses 14% of explainable disease risk. Evaluating physical measurements alongside blood lipid chemistry measurements increases risk identification accuracy to nearly 50%. Consequently, Health services can integrate composite clinical scoring (combining BMI, BP, and s5) into their electronic health record system to facilitate identifying high-risk patients.

Alternatively, controlling more measurements such as activity level, prescribed medications, genetic background, etc. and redoing the analysis can help finding confounding measurements, and as a result, a more reliable model for statistically suggested relationships or an early-alarm workflow.

Even if significant, we can not link the cause of the increase or decrease in progression to the given features, since the accuracy is far from perfect. But the three mentioned features can help the medical staff with their duties, especially if subjected to more data and model improvements.

High statistical certainty at the population level does not eliminate individual patient variability. Clinical management must pair predictive risk scores with individualized diagnostic evaluations and always have the outcome controlled by the medical staff.

[The complete slide presentation asset tailored specifically for non-specialist decision-makers is present within the project repository.]

8. Conclusion and Reflection

An Analytical workflow and an adequate approach to data was a crucial step for this, and any other analysis, since data is the heart of this type of studies and the captain to the ship, being our model. A major insight was recognizing and understanding the distinction between statistical significance and practical importance; a highly significant never guarantees a certain cause, especially in the medical field. A model's parameters may be statistically significant, but the results being poor and unusable.

Noticing the inter-predictor collinearity among serum measurements presented a significant insight as one approach to avoid multicollinearity. Biochemical knowledge, to a certain level, was required for better understanding the data and the objective at hand. Summarizing a statistical finding to a different group and translating the findings into a different language for a different field was indeed a rewarding challenge.

Given additional time and computational resources, three key improvements would be suggested: (1) Implementing non-linear modeling techniques to capture potential non-linear threshold effects; (2) Utilizing k-fold cross-validation to assess out-of-sample predictive performance; and (3) Benefitting more predictive features that can assist making better predictions and a stronger model, collected across a larger sample.